

ZENELAIT LMS

COMPREHENSIVE ROLE-BASED LEARNING MANAGEMENT MANUAL

An all-inclusive, multi-role digital learning platform designed for modern educational institutions and training academies. Built by Zenelait InfoTech Private Limited to streamline workflows from classroom attendance to board-level financial and academic audits.

CORPORATE PROFILE

Zenelait InfoTech is a leading software engineering firm dedicated to building next-generation enterprise applications. We specialize in designing customizable SaaS platforms that integrate CRM, ERP, and learning management networks into unified hubs.

ZENELAIT LMS VISION

Zenelait LMS enables educational institutions to achieve 100% operational transparency. Our platform bridges the gap between daily classroom metrics and executive administration auditing, delivering automated compliance checks, integrated payment gateways, and AI-driven student performance insights.

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Security Note: User role configurations are validated dynamically at database query compile-time. Any route modifications are blocked by client side Auth guards.

1. SYSTEM ARCHITECTURE & TECHNICAL STACK

Zenelait LMS employs a highly responsive, decoupled client-server architecture model. The client-side dashboard framework runs on a React Single Page Application (SPA), connecting to a multi-threaded Spring Boot 3 enterprise application that orchestrates business logic and communicates directly with external cloud-based Generative AI models.

TECHNICAL STACK DETAILS

Layer	Technology	Responsibility
Frontend Client	React 18, Styled-Components, Recharts	Role Dashboards, Real-time Charts, Responsive Layouts, Client-side routing, Export Utilities.
Java Backend	Spring Boot 3.2, Java 17/20, Hibernate JPA, Apache Maven	Business Logic, Multi-role REST Data APIs, Security Filter Chains, and Batch Scheduling.
Generative AI Integration	Google Gemini API (gemini-2.5-flash), Hugging Face Inference API	Intelligent Role-Specific Assistant (student performance, parent audits, attendance, and fee querying).
Database	MySQL 8.x	Relational Schemas, User Ledgers (single-table User model with polymorphic roles), Batches/Courses Catalog, Fee Logs.

SECURITY FILTER CHAIN & DECOUPLING

All network interactions use JSON-encoded requests. Spring Security handles endpoint protection, verifying JWT signature keys encoded in client-header HTTP bearer authentication tokens (Authorization: Bearer <accessToken>) on every dispatch.

The backend employs database connection pools (HikariCP) to interact with relational schemas, ensuring fast transactional speeds during heavy academic logs updates, attendance submissions, and fee updates.

2. ACCESS SECURITY & AUTHENTICATION FLOW

Access control starts at the login gateway. On success, authentication endpoints return JSON-encoded JWT tokens containing role mappings which dictate access permissions across the system.

- **Auth Registration:** Onboards student, teacher, parent, and admin roles. Admins require secondary verification values (e.g., referralId).
- **Auth Login:** Verifies credentials against BCrypt (strength 10) password hashes. Emits JWT access (24h TTL) and refresh (7 days TTL) tokens.

3. ADMIN PORTAL (SYSTEM CONFIGS)

The Admin Portal is the dashboard for system configuration, reporting statistics, and platform maintenance.

WHAT THE ADMIN SEES:

- **Dashboard Stats:** Aggregated system data showing total count of students, teachers, parents, and courses.
- **User Directories:** Polished tables filtering and indexing users by role, code, and active/inactive status.
- **System Log Health:** Direct access logs indicating performance latency and active connections.

WHAT THE ADMIN POSTS:

- **User Registrations:** Activates, suspends, or deletes user accounts.
- **Course Catalog:** Adds new courses, defining duration hours and assigning teachers.
- **Batches Configurator:** Groups students into active batches associated with specific course timelines.
- **Tuition Invoicing:** Assigns fee records with due dates directly to student profiles.

4. TEACHER PORTAL (ACADEMIC CONTROLS)

Designed for educators, the Teacher Portal streamlines daily academic logs, grading tasks, and classroom management.

WHAT THE TEACHER SEES:

- **My Courses Catalog:** Dynamic grid showing courses currently assigned to their profile.
- **Submission Inbox:** Submissions uploaded by students, queued for grading.
- **Timetable Schedules:** Mapped classroom hours and lecture slots.

WHAT THE TEACHER POSTS:

- **Assignment Publisher:** Sets homework (instructions, due dates, maximum marks).
- **Grading & Feedback:** Logs marks and text assessments onto student submissions.
- **Attendance Register:** Logs student classroom statuses (PRESENT, ABSENT, LATE).

5. STUDENT PORTAL (LEARNING DASHBOARD)

The Student Portal gives learners structured visibility into their classes, assignments, and grades.

WHAT THE STUDENT SEES:

- **Academic Progress Board:** Visual progress charts showing grade curves and attendance rates.
- **My Assignments:** Deadlines, descriptions, and grading status of tasks.
- **Fee Ledger:** Summaries of unpaid and paid tuition records.

WHAT THE STUDENT POSTS:

- **Course Enrollment Requests:** Submits requests to join specific course batches.
- **Coursework Uploads:** Submits files or text answers for evaluation.

6. PARENT PORTAL & INTEGRATED WALLET

The Parent Portal provides parents with academic visibility and tuition fee payment channels.

WHAT THE PARENT SEES:

- **Child Academic Audit:** Real-time access to linked children's attendance histories and grading trends.
- **Invoicing Index:** Complete ledger of pending, due, and paid tuition fees.
- **Announcements Feed:** Broadcaster notifications targeted specifically to parent profiles.

WALLET & RAZORPAY INTEGRATION:

To simplify tuition payments, the platform utilizes a Parent Wallet schema linked to the Razorpay Payment Gateway API.

- **Wallet Top-ups:** Parents securely load funds into their wallet using card payments, net banking, or UPI links.
- **Fee Settlements:** Parents pay pending tuition bills directly using their wallet balance. Funds are transferred and records are immediately marked as PAID in the database.



7. AI ASSISTANT & INSIGHTS PIPELINE

Zenelait LMS features direct API integration with Google Gemini 2.5 Flash and Hugging Face Inference servers to provide contextual assistance based on active database records.

- **Gemini 2.5 Flash Pipeline:** Handles text generation prompts, fetching active metrics (grades, attendance, profiles) to contextualize queries.
- **Role-Specific Insights:**
 - *Student:* Analyzes grade records to formulate personalized study suggestions.
 - *Parent:* Evaluates attendance data and warns of potential academic risks.
 - *Teacher:* Auto-generates template comments for grading reviews.

8. AUTOMATED COURSE CERTIFICATES

Once a student successfully completes all batch requirements, the system unlocks certificate generation.

- **Dynamic Canvas Rendering:** Extracts completion data (student name, batch duration, course title) and compiles it using html2canvas and jspdf into verified PDF certificates.
- **Unique Verification Hash:** Each certificate embeds a verification code referencing records in the database, allowing third parties to validate credentials online.